```
      insert(E, (-1.5, 2), 2)
     insert(null, (-1.5, 2), 3)
     return new KdNode((-1.5, 2), null, null, 3)
    return E
  return B
return A
```

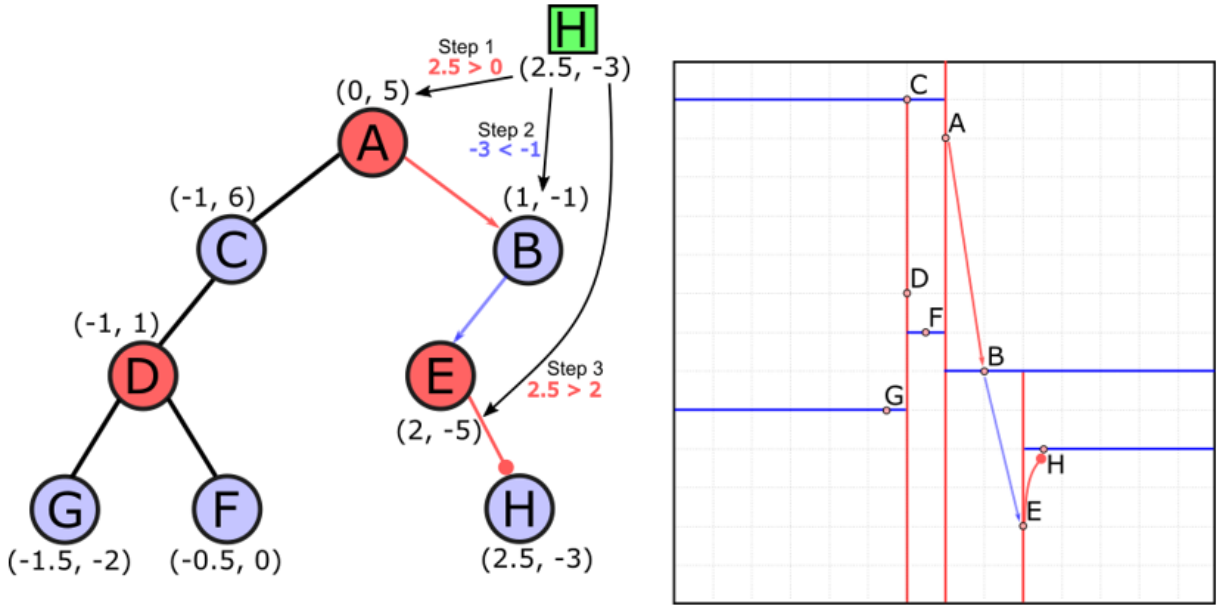


Figure 9.12 Another example of inserting a new point on a k-d tree (2-D). The starting point is the k-d tree resulting after the insertion in figure 9.11.

9.3.3 Balanced Tree

Before moving on to the most advanced methods designed for k-d trees, let's take a step back. In section 9.2.3 we have already described one key point of k-d trees: we need our tree to be balanced. In the previous sections we have indeed seen how search and insertion have running time $O(h)$, proportional to the height of the tree, and thus having a